

## PHASE

# 14

## Userland: libc, Shell and Utilities

Build the user-space half of the system: a C library, pipes, a real terminal, the Turk shell, and the everyday commands

Overall progress toward Turk-OS 1.0 (this phase in red)

90% → 96%

### 🕒 Duration

3 weeks

### ★ Difficulty

★★★★☆ (4/5)

### ✂ You need

Phase 13: VFS, descriptors, TurkFS; Phase 11: fork/exec/wait, signals

### 🔧 You will have

Turk-libc (system call wrappers with `errno`, `strings`, `malloc`, buffered `stdio` with `printf`), a pipe system call, a terminal line discipline with `echo`, line editing, `Ctrl+C` and `Ctrl+D`, basic ANSI escape codes on the console, `tsh` (the Turk shell) with pipelines, redirection and background jobs, about fifteen core utilities, and a boot that ends at a shell prompt.

## 1 Why this phase

Until now you have been talking to Turk-OS through a monitor built into the kernel. A real UNIX has no such thing: the shell is an ordinary user program, and so are `ls`, `cat` and everything else you type. That design is only possible because of the system calls you built in Phases 10–13, and this phase is where you find out whether they really work together. **OSTEP** chapter 5 argues that `fork`, `exec`, `wait` and file descriptors are exactly the right primitives for a shell; now you get to confirm it by writing one.

Most of the code in this phase runs in ring 3. The kernel gains two things: **pipes**, which connect one process's output to another's input, and a proper **terminal** that edits lines and turns `Ctrl+C` into a signal. Everything else is user space: a C library so programs can be written normally, the shell, and the utilities. When you finish, Turk-OS boots to a prompt, and you can type `ls /bin | wc -l > count.txt`.

## 2 Concepts you need

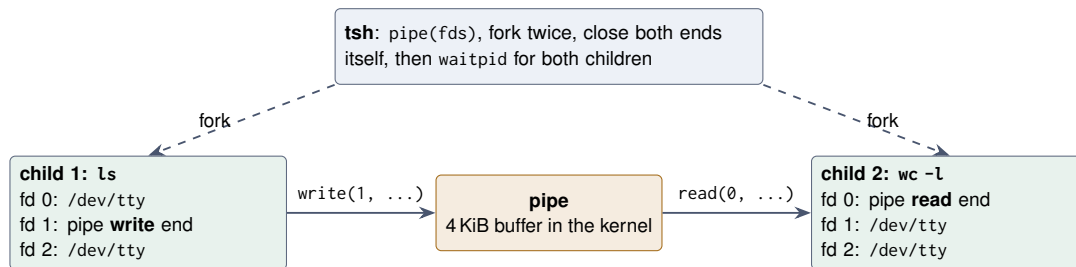
### 2.1 The C library

User programs don't make system calls directly; they call the **C library**, which provides the standard interface (`string.h`, `stdio.h`, `stdlib.h`, `unistd.h`, ...) on top of system calls. Some functions are thin wrappers: `write` puts its arguments in registers and executes `int 0x80`. Others do real work in user space: `printf` formats into a buffer and calls `write` only when the buffer fills or a line ends, and `malloc` manages a heap that it grows with `brk`, reusing your Phase 7 allocator almost unchanged. System calls return a negative error code (`-ENOENT`); by convention the library stores the positive code in the global `errno` and returns `-1`, which is what programs expect.

### 2.2 Pipes and redirection

A **pipe** is a kernel buffer with two file descriptors: whatever is written to one end can be read from the other. The shell builds a pipeline with the `fork/exec` gap that **OSTEP** chapter 5 describes: it creates a pipe, forks two children, and in each child uses `dup2` to move the right end of the pipe onto descriptor 0 or 1 before calling `exec` (Figure 1). The programs never know. Redirection (`> file`) is the same trick with a file instead of a pipe. A reader sees end-of-file only when *every* write end is

closed, so every process, the shell included, must close the ends it doesn't use. Forgetting that is the classic pipe bug: the pipeline never finishes.



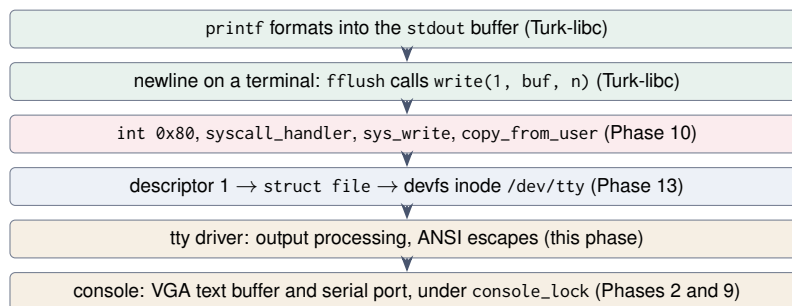
**Figure 1:** How `ls | wc -l` is wired. Each child dup2s its pipe end onto fd 0 or 1 and closes the originals before exec.

## 2.3 The terminal line discipline

When you type at a UNIX prompt, the program doesn't see each key. The kernel's terminal driver collects a whole line, echoes characters, handles Backspace (erase) and Ctrl+U (kill the line), and hands the line over only when you press Enter. This is **canonical** (cooked) mode. Special keys become actions: Ctrl+D at the start of a line means end-of-file, and Ctrl+C sends SIGINT to the **foreground** process instead of being passed as input. Programs like editors switch the terminal to **raw** mode to get every key at once. **OSDI** §3.8.2 describes all of this, and **MOS** §5.6 covers the output side, including the escape sequences (such as ESC [2J to clear the screen) that let programs move the cursor and change colours.

## 2.4 Every layer at once

This is the phase where a single `printf("merhaba\n")` passes through almost everything you have built (Figure 2). When something prints the wrong thing, walk down this list and check each layer on its own.



**Figure 2:** The path of one `printf` in Turk-OS.

## 3 Reading plan

| Priority           | Book         | What to read                                                                                             | Pages | Why now                                                       |
|--------------------|--------------|----------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------|
| <b>REQUIRED</b>    | <b>OSTEP</b> | Ch. 5 <i>Interlude: Process API</i> ; re-read §5.4 on why fork/exec suits a shell, redirection and pipes | 37–48 | The design of tsh in a few pages.                             |
| <b>REQUIRED</b>    | <b>LOB</b>   | Ch. 11, sections <i>Using C for User Mode Programs</i> and <i>A C Library</i>                            | 67–68 | What a minimal libc needs.                                    |
| <b>RECOMMENDED</b> | <b>OSDI</b>  | §1.3.3 <i>The Shell</i> ; §1.4 <i>System Calls</i>                                                       | 25–42 | The POSIX calls your library wraps, with a sketch of a shell. |

| Priority    | Book | What to read                                                                                             | Pages            | Why now                                                            |
|-------------|------|----------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------|
| RECOMMENDED | OSDI | §5.6.9 <i>Pipes and Special Files</i>                                                                    | 563–565          | How MINIX implements pipes and device files.                       |
| RECOMMENDED | OSDI | §3.8.2 <i>Terminal Software</i> ; §3.8.4 <i>Implementation of the Device-Independent Terminal Driver</i> | 307–316, 331–350 | Canonical mode, special characters, and a working line discipline. |
| RECOMMENDED | MOS  | §5.6.1–5.6.2 <i>Input Software, Output Software</i>                                                      | 394–405          | Cooked versus raw input, escape sequences.                         |
| RECOMMENDED | OSC  | §2.1–2.2 <i>Services, User and OS Interface</i> ; §2.4 <i>System Services</i>                            | 55–62, 74–75     | Where shells and utilities fit.                                    |
| OPTIONAL    | MOS  | §1.5.6 <i>The Shell</i> ; §10.2 <i>Overview of Linux</i> (interfaces, shell, utility programs)           | 45–46, 723–733   | The user’s view of a UNIX system.                                  |
| OPTIONAL    | MOS  | §12.2 <i>Interface Design</i>                                                                            | 985–993          | Keep the library and shell simple and orthogonal.                  |

### Q Real-system lens: how MINIX 3 does it

The MINIX terminal driver, `drivers/tty/tty.c` (13600), implements canonical mode in `in_process`: it checks every incoming character against the special characters in the terminal’s settings (erase, kill, EOF, interrupt, ...), edits the input queue, echoes, and completes a pending read when a line is ready. Pipes live in the file-system server, `servers/fs/pipe.c` (25900): a pipe is an inode whose data blocks act as the buffer, and `suspend/release` put readers and writers to sleep and wake them, just as your wait queues will.

## 4 Build it

### Task 14.1 Turk-libc: the skeleton

Create `user/libc/` with headers in `user/include/`, built into a static `libc.a` with `i686-elf-ar`. Every program links `crt0.o`, its own objects and `libc.a`. Start with system call wrappers that follow the `errno` convention, then `string.h` and `ctype.h` (copy them from the kernel’s `lib/`; you wrote them in Phase 0).

`user/libc/unistd.c`

```
int errno;

static int ret_errno(int32_t r) /* kernel returns -E...; libc returns -1 */
{
    if (r < 0) {
        errno = -r;
        return -1;
    }
    return r;
}

int read(int fd, void *buf, size_t n) { return ret_errno(syscall3(SYS_read, fd, (uint32_t)buf, n)); }
int write(int fd, const void *buf, size_t n) { return ret_errno(syscall3(SYS_write, fd, (uint32_t)buf, n)); }
↵
int open(const char *path, int flags) { return ret_errno(syscall3(SYS_open, (uint32_t)path, flags, 0)); }
int pipe(int fds[2]) { return ret_errno(syscall3(SYS_pipe, (uint32_t)fds, 0, 0)); }
```

### Task 14.2 malloc in user space

Port your Phase 7 heap: replace `ksbrk` with the `brk` system call and `ASSERT` with an `abort()` that prints a message and exits with `SIGABRT`. Add `calloc`, `realloc` and `free`. Because `brk` allocates lazily (Phase 11), a program that asks for a large heap but touches little of it stays cheap.

✓ **Verify:** The heap stress test from Phase 7, compiled as a user program, passes.

### Task 14.3 stdio

Implement `FILE` with a buffer, a mode (unbuffered, line-buffered, fully buffered) and a descriptor, plus `stdin`, `stdout` and `stderr`. `stdout` is line-buffered when it is a terminal (check with an `isatty` built on `fstat`) and fully buffered otherwise; `stderr` is unbuffered. Port your `kvprintf` engine for `printf`, `fprintf`, `sprintf` and `snprintf`, and add `fopen`, `fclose`, `fgets`, `fputs`, `fread`, `fwrite`, `fflush`, `getchar` and `putchar`. Make `exit` flush all open `FILE`s before calling the system call, while `_exit` doesn't.

✓ **Verify:** `printf` output appears without an explicit `fflush` when printing to the terminal, and is complete in a file even when the program ends with `return 0` from `main`.

### Task 14.4 Pipes in the kernel

Add `sys_pipe`: create a pipe object with a 4 KiB ring buffer, counts of open readers and writers, a spinlock and two wait queues, then two `struct files` (read end and write end) and two descriptors. A read from an empty pipe sleeps until data arrives, or returns 0 (end-of-file) if there are no writers left. A write to a full pipe sleeps until there is room; if no readers are left, it sends `SIGPIPE` to the writer and returns `-EPIPE`. Closing an end decrements its count and wakes the other side.

kernel/fs/pipe.c

```
#define PIPE_SIZE 4096

struct pipe {
    uint8_t      buf[PIPE_SIZE];
    uint32_t      rpos, wpos;          /* bytes in pipe = wpos - rpos */
    int           readers, writers;    /* open ends */
    spinlock_t    lock;
    struct wait_queue can_read, can_write;
};

int32_t pipe_read(struct pipe *p, uint8_t *dst, uint32_t n)
{
    spin_lock(&p->lock);
    while (p->wpos == p->rpos) {        /* empty */
        if (p->writers == 0) {         /* nobody can ever write again */
            spin_unlock(&p->lock);
            return 0;                  /* end-of-file */
        }
        sleep_on(&p->can_read, &p->lock);
    }
    uint32_t got = 0;
    while (got < n && p->rpos != p->wpos)
        dst[got++] = p->buf[p->rpos++ % PIPE_SIZE];
    wake_up_all(&p->can_write);
    spin_unlock(&p->lock);
    return got;
}
```

The loop copies into a kernel buffer; `sys_read` then hands it to the user with `copy_to_user` after the lock is released, because `copy_to_user` can fault and sleep.

✓ **Verify:** A test program writes 1 MiB into a pipe from a child and reads it back in the parent, checking every byte; the writer blocks whenever the buffer is full.

### Task 14.5 A real terminal

Turn `/dev/tty` into a proper terminal driver. Keyboard characters go into an input queue. In canonical mode, echo them; Backspace removes the last character from the current line and from the screen; Ctrl+U removes the whole line; Enter completes the line and wakes readers; Ctrl+D on an empty line makes the pending read return 0; Ctrl+C sends SIGINT to the foreground process group and discards the line. Add an `ioctl` (number 54) so a program can read and set a flags word (canonical on or off, echo on or off) and set the foreground process group, which the shell does before running a job and after it finishes. On the output side, teach the console the few ANSI sequences programs need: ESC [2J (clear), ESC [H and ESC [row;colH (move the cursor), ESC [K (clear to end of line) and ESC [...m (colours).

✔ **Verify:** A program that reads lines and prints them in capitals works like a normal UNIX filter: editing works, Ctrl+D ends it, Ctrl+C interrupts it.

### Task 14.6 tsh, the Turk shell

Write `user/sh/tsh.c`. Read a line, split it into words (honouring single and double quotes), and parse it into one or more pipelines separated by `;`. Each pipeline is a list of commands, each with its own redirections (`<`, `>`, `>>`), plus an optional `&` for background. Built-ins run inside the shell because they must change its own state: `cd`, `pwd`, `exit`, `export` and `help`. Everything else runs with `fork` and `exec`, looked up in the directories of `PATH` (default `/bin`). Before each prompt, reap finished background jobs with `waitpid(-1, &status, WNOHANG)`. The shell ignores SIGINT itself and makes each foreground job the terminal's foreground group, so Ctrl+C reaches only the job. The core of the pipeline code looks like this:

user/sh/tsh.c (running a pipeline)

```
static void run_pipeline(struct cmd *cmds, int n, bool background)
{
    int prev_read = -1;
    pid_t pids[MAX_CMDS];

    for (int i = 0; i < n; i++) {
        int fds[2] = { -1, -1 };
        if (i < n - 1 && pipe(fds) < 0) { perror("pipe"); return; }

        pid_t pid = fork();
        if (pid == 0) { /* child */
            if (prev_read != -1) { dup2(prev_read, 0); close(prev_read); }
            if (i < n - 1) { dup2(fds[1], 1); close(fds[0]); close(fds[1]); }
            apply_redirections(&cmds[i]); /* <, >, >> override the pipe */
            execvp(cmds[i].argv[0], cmds[i].argv);
            perror(cmds[i].argv[0]);
            _exit(127); /* exec failed */
        }
        pids[i] = pid;
        if (prev_read != -1) close(prev_read); /* the shell keeps no pipe ends */
        if (i < n - 1) { close(fds[1]); prev_read = fds[0]; }
    }
    if (!background)
        for (int i = 0; i < n; i++)
            waitpid(pids[i], NULL, 0);
}
```

✔ **Verify:** `echo merhaba | wc -c` prints 8; `ls /bin > /mnt/list.txt` followed by `cat < /mnt/list.txt` shows the listing; `sleep 5 &` returns to the prompt at once and is reaped later.

### Task 14.7 Core utilities

Write each as a small program in `user/bin/`. Keep them simple and predictable; they double as tests for the kernel.

| Group     | Programs and what they exercise                                                                                                                                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| text      | <code>echo</code> , <code>cat</code> (files or <code>stdin</code> ), <code>wc</code> ( <code>-l -w -c</code> ), <code>hexdump</code> : read, write, pipes                                                                                  |
| files     | <code>ls</code> (with <code>-l</code> ), <code>mkdir</code> , <code>rm</code> , <code>cp</code> , <code>mv</code> , <code>touch</code> : <code>getdents</code> , <code>stat</code> , open flags, <code>unlink</code> , <code>rename</code> |
| processes | <code>ps</code> , <code>kill</code> , <code>sleep</code> : a <code>getprocs</code> system call that copies a table of task information to the user, signals, timers                                                                        |
| system    | <code>uptime</code> , <code>clear</code> (ANSI escape), <code>sync</code> , <code>shutdown</code> ( <code>sync</code> , then power off QEMU with <code>outw(0x604, 0x2000)</code> in the kernel)                                           |

### Task 14.8 Boot into the shell

Make `init` open `/dev/tty` for descriptors 0–2, print a banner, start `tsh`, and start a new one whenever it exits, while still reaping orphans. Keep the old kernel monitor reachable for emergencies, for example on the serial port or with a monitor boot option. Code page 437 has no `$`, so a banner like “Turk-OS 1.0 – Hos geldiniz!” stays ASCII until Phase 15.

✓ **Verify:** Turk-OS boots straight to `turk@os:/$`, and typing `exit` brings back a fresh shell.

### Task 14.9 Test the userland

Add a `-c` option to `tsh` (run one command line and exit) and a test runner that executes a list of command lines and compares their output with expected text. Put these cases in `make test`.

| Command line                                           | Expected output                          |
|--------------------------------------------------------|------------------------------------------|
| <code>echo a b c</code>                                | <code>a b c</code>                       |
| <code>echo merhaba   wc -c</code>                      | <code>8</code>                           |
| <code>echo x &gt; /mnt/t; cat /mnt/t; rm /mnt/t</code> | <code>x</code>                           |
| <code>cat /etc/motd   cat   cat   wc -l</code>         | the number of lines in <code>motd</code> |
| <code>nosuchprogram</code>                             | an error message, exit status 127        |

## 5 Testing and debugging

| Symptom                                                   | Likely cause                                                                                      | What to do                                                                                                                                     |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| A pipeline never finishes                                 | Some process (often the shell) still holds a pipe write end, so the reader never sees end-of-file | Close every unused end in every process; list open descriptors with a debug command.                                                           |
| Output from <code>printf</code> is missing                | Buffered data never flushed, because the program called <code>_exit</code> or crashed             | <code>exit</code> must flush; use <code>stderr</code> for debug output.                                                                        |
| <code>Ctrl+C</code> kills the shell                       | The shell didn't ignore <code>SIGINT</code> , or it stayed the foreground group                   | Ignore <code>SIGINT</code> in <code>tsh</code> ; set the job as foreground before <code>exec</code> and take it back after <code>wait</code> . |
| <code>ls</code> in a pipe prints nothing, but works alone | The program writes to the terminal directly instead of to descriptor 1                            | All output goes through <code>write(1, ...)</code> ; only the <code>tty</code> driver knows about the screen.                                  |
| Zombies accumulate after background jobs                  | Background children are never waited for                                                          | Reap with <code>WNOHANG</code> before every prompt.                                                                                            |

| Symptom                                       | Likely cause                                                                   | What to do                                          |
|-----------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------|
| Arguments arrive garbled in <code>main</code> | Wrong <code>argv</code> layout on the initial stack or missing NULL terminator | Compare the stack with the Phase 11 diagram in GDB. |

## 6 Milestone

### Milestone: A usable system

- ☐ Turk-libc provides system call wrappers with `errno`, `strings`, `malloc` and buffered `stdio`.
- ☐ Pipes, the terminal line discipline and ANSI output work.
- ☐ Turk-OS boots to a `tsh` prompt; pipelines, redirection, background jobs and `Ctrl+C` work.
- ☐ The core utilities are present and pass the scripted tests.
- ☐ No zombies remain after a long interactive session.

### Self-check questions

1. Why does the shell fork before `exec`? What would break if it called `exec` directly?
2. Explain exactly how `ls | wc` is wired with pipe, fork and `dup2`.
3. Why must unused pipe ends be closed, and by whom?
4. Why is `cd` a built-in and not a program in `/bin`?
5. Canonical versus raw mode: who implements line editing in Turk-OS, and why not the shell?
6. List every layer a `printf` passes through on its way to the screen.

## 7 Going further

Add job control: process groups, `fg`, `bg` and jobs, and `Ctrl+Z` sending `SIGTSTP`. Give `tsh` variables, `if`, `for` and scripts run from files. Write a small full-screen text editor that uses raw mode and ANSI escapes. Or port an existing program with modest needs; a small interpreter such as Lua is a classic target, and the missing libc functions it asks for will show you what your library still lacks.

Next: **Phase 15 Hardening, Polish and Release: Turk-OS 1.0** →

### Books used in this plan

- LOB** Erik Helin, Adam Renberg. *The little book about OS development*, 2015. The hands-on x86 guide. Free at <https://littleosbook.github.io>.
- OSTEP** Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau. *Operating Systems: Three Easy Pieces*, v0.90, Arpaci-Dusseau Books, 2015. Main theory text. Free at <https://pages.cs.wisc.edu/~remzi/OSTEP/>.
- OSC** A. Silberschatz, P.B. Galvin, G. Gagne. *Operating System Concepts*, 10th ed., Wiley, 2018. Reference theory text.
- MOS** Andrew S. Tanenbaum, Herbert Bos. *Modern Operating Systems*, 4th ed., Pearson, 2015. Hardware, I/O, design, security.
- OSDI** Andrew S. Tanenbaum, Albert S. Woodhull. *Operating Systems: Design and Implementation*, 3rd ed., Pearson Prentice Hall, 2006. MINIX 3 with full source listing.

The books do not come with this plan. **LOB** and **OSTEP** are free online; buy or borrow the other three (the roadmap lists their ISBNs). Page numbers are the *printed* page numbers of these editions, not PDF page indices. With another edition, or with the free **OSTEP** chapters (each one starts at page 1), find the reading by its chapter and section numbers. MINIX source references like `kernel/proc.c` (07400) give the file and its starting line in **OSDI** Appendix B.